Amplicon library workflow: two-step PCR or one-pot PCR

both routes give the same barcoded molecule and share QC, pooling, purification and Evoprep · HuangLab LIMS stages 1 → 3a → 3b → 4a

Stage 1 — Specimen# / Culture#

Stage 3a — DNA extraction · MolWork#

Stage 3b — PCR + gel + Qubit · pcr_reaction row

Stage 4a — Nanopore experiment · Nanopore#

Specimen registered in LIMS
soil / environmental → Specimen# | fungal strain → Culture#

Soil DNA extraction
SuperSoil protocol

Fungal DNA extraction
modified BOMB v9.0

DNA extraction

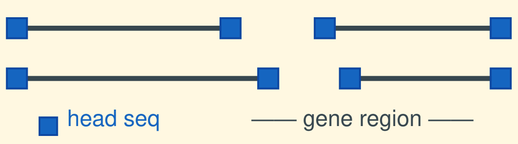
DNA template → MolWork#
soil crude extract (SuperSoil) | pure fungal extraction (BOMB v9)
MolWork# minted after quantification

choose a route

Route A · two-step PCR

1st PCR (gene regions)

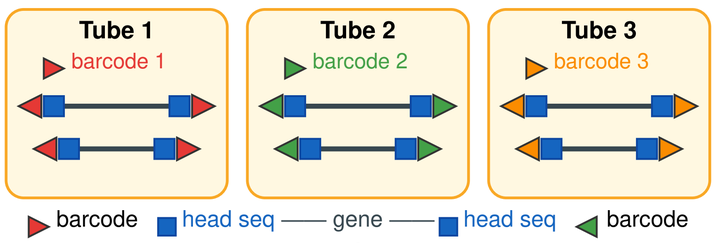
1st PCR products (per sample)



2nd PCR (barcoding)
gel check #1: good amplicons only

2-step PCR protocol
mixtures + cycling

2nd PCR products: one tube per sample, one barcode per tube



Route B · one-pot PCR

Set up one-pot PCR

One reaction tube (per sample)

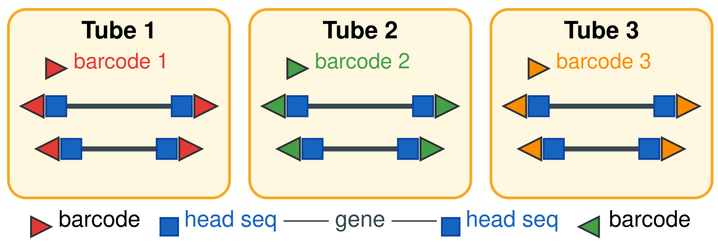
gene-region primers (head-tailed, 0.1 µM each) + **barcode primers** (barcode + head seq, 0.3 µM each)

50 µL: KAPA HiFi 2X + both primer sets + 3 µL template

One-pot PCR
phase 1: 25 cycles, 60–66 °C
phase 2: 15 cycles, 65 °C

One-pot PCR protocol
50 µL mix + 25/15 cycles

One-pot PCR products: one tube per sample, one barcode per tube



same molecule from either route



Gel check → Qubit Flex (Stage 3b)
confirm barcoded product, quantify → 10 ng/µL



LIMS register
`/lims-register`
Stage 3b row: PCR_<Ta> or one_pot_<Ta>

Pooling (equal molar)

Pool all barcoded samples
equal molar of every sample

Pooled sample

Purification (bead clean-up)
PCR sample : bead = 1 : 1

Size selection 1X
bead purification protocol

Purified pooled sample

Ready to be sequenced
library construction using Evoprep

EvoPrep library prep
KAPA EvoPrep NBD114.96

Legend

- Head seq (universal tail on the gene-region primers)
- Barcode (sample-specific, on the barcode primers)
- Gene region (target amplicon; colour = different sample)

QR codes (left) open the lab's self-hosted protocol PDF for that step
left bands = HuangLab LIMS stages (/lims-register); dashed = QC / info flow
Route A: extra gel check after the 1st PCR · Route B: none